

EPSY 5244: Survey Design, Sampling, and Implementation

Wenchao Ma
University of Minnesota



Concepts

- A concept is an idea, a general mental formulation summarizing specific occurrences
- Objective concepts
 - Often “observable”
 - Can be quantified in some objective way
 - Examples: Age, academic major, years of teaching experience



Concepts

- A concept is an idea, a general mental formulation summarizing specific occurrences
- Subjective concepts
 - Also referred to as “(psychological) constructs”
 - an underlying, unobservable personal attributes, traits, or characteristics of an individual
 - a label for a cluster or domain of related behaviors
 - often assumed to be an underlying cause of a cluster or domain of related behaviors
 - are hypothetical
 - Examples: Happiness, love, satisfaction, self esteem, agreeableness



- Concepts must be defined



Concepts

- How is “Unemployment” defined?
- People are classified as unemployed if they do not have a job, have actively looked for work in the prior 4 weeks, and are currently available for work.
- Actively looking for work may consist of any of the following activities:
 - Contacting an employer directly or having a job interview
 - Submitting resumes or filling out applications
 - Placing or answering job advertisements
 - Checking union or professional registers
 - Some other means of active job search



Concepts

- Passive methods of job search do not have the potential to connect job seekers with potential employers and therefore do not qualify as active job search methods.
- Examples of passive methods include attending a job training program or course, or merely reading about job openings that are posted in newspapers or on the Internet.



Concepts

- Workers expecting to be recalled from temporary layoff are counted as unemployed whether or not they have engaged in a specific job seeking activity.
- In all other cases, the individual must have been engaged in at least one active job search activity in the 4 weeks preceding the interview and be available for work (except for temporary illness).

Source: https://www.bls.gov/cps/cps_htgm.htm#unemployed



Concepts

- How do we define “self-esteem?”
- Self-esteem is defined as
 - a positive or negative orientation toward oneself
 - an overall evaluation of one's worth or value.
 - People are motivated to have high self-esteem, and having it indicates positive self-regard, not egotism.
 - Self-esteem is only one component of the self-concept



Concepts

- Concepts must be operationalized

To measure unemployment, can we directly ask people if they are unemployed?

To measure self-esteem, can we directly ask people their self-esteem level?



Operationalization of Unemployment

- Does anyone in this household have a business or a farm?
- Last week, did you do any work for (either) pay (or profit)?
- If the answer to question 1 is "yes" and the answer to question 2 is "no," the next question is:
- Last week, did you do any unpaid work in the family business or farm?
- For those who reply "no" to both questions 2 and 3, the next key questions used to determine employment status are:
- Last week, (in addition to the business) did you have a job, either full or part time? Include any job from which you were temporarily absent.
- ...

Source: https://www.bls.gov/cps/cps_htgm.htm#unemployed



Operationalization of Self-Esteem

- <https://www.norton.com/college/psych/psychsci/media/rosenberg.htm>



Measurement

- Assignment of numerals to objects or events according to rules
- When measuring construct, one assigns numbers to individuals
- The number assigned represents the level of the construct for the individual
- Four levels
 - Nominal
 - Ordinal
 - Interval
 - Ratio



Measurement - Nominal

- Different numbers represent distinguish different categories
- Numbers do not have any real numerical meaning
- It is not appropriate to add, subtract, or otherwise manipulate them numerically.



Measurement - Ordinal

- Numbers represent a rank order of the properties of objects
- Although the numbers used in ordinal measurement imply rank order, the intervals between adjacent scale points are not assumed to be equal
- Ordinal measures can be transformed in any way that preserves the original order



Measurement - Interval

- No true zero
- Appropriate to calculate nearly all parametric statistics, such as the mean, standard deviation, and correlation from interval level data



Measurement - Ratio

- Distinctiveness, order, equal intervals and true zero point
- A true zero point is one that represents the absolute lack of the property being measured



- What measurement scale are “Unemployment” and “Self-Esteem” on?



Where can you find psychological tests?

- Mental measurements yearbook (MMY) can be accessed from UofM library.

Self-Esteem Assessment

Acronym:	SEA
Authors:	PsychTests AIM, Inc.
Publication Date:	2011.
Publisher Information:	PsychTests AIM, Inc., 1558 rue Viel, Suite 103, Montreal, Quebec, H3M 1G5, Canada, ilona@psychtests.com , https://archprofile.com/corporate
Source:	J. F. Carlson, K. F. Geisinger, & J. L. Jonson (Eds.), The nineteenth mental measurements yearbook. 2014.
Reviewers:	Buhs, Eric S. ; Scott, Amy Nicole
Purpose:	'Designed to evaluate an individual's general level of self-esteem .'
Test Category:	Personality.
Population:	Under age 17 through adult.
Scores:	9: Feelings of Inadequacy, Sense of Self -Worth, Need for Approval, Unrealistic Self -Expectations, Sense of Social Acceptance, Narcissism, Defensiveness, Self -Deception, Overall Score.
Administration:	Individual.
Time:	(30) minutes.
Price Data:	Available from publisher.
Comments:	Self -administered online assessment. The test publisher advises that the test manual is being updated to include more information about methodology and theoretical background used in the development of the test. The test publisher also advises that this information is available to clients, as are benchmarks for relevant industries and racial/ethnic group comparison data. However, this information was not provide to Buros or the reviewers
Cross References:	For reviews by Eric S. Buhs and Amy Scott, see 19.149 .
Published Test Description:	Self-Esteem Assessment. Purpose: "Designed to evaluate an individual's general level of self-esteem ." Population: Under age 17 through adult. Publication Date: 2011. Acronym: SEA. Scores: 9: Feelings of Inadequacy, Sense of Self -Worth, Need for Approval, Unrealistic Self -Expectations, Sense of Social Acceptance, Narcissism, Defensiveness, Self -Deception, Overall Score. Administration: Individual. Price Data: Available from publisher. Time: (30) minutes. Comments: Self -administered online assessment. The test publisher advises that the test manual is being updated to include more

